ANALYSTLAB AFRICA EXPERIENCE LAB · WEEK 8 · BATCH E

HealthConnect: Final ML Pipeline Integration & Readiness

Gyimah Ramsey Opoku

Machine Learning Engineering track

INTRODUCTION

The problem and my role

Project question: How can HealthConnect Clinic use data and AI to reduce missed appointments and improve the patient support experience?

The gap I own

- A model in a notebook can't be used by anyone else
- Other tracks and a clinic system need to score patients the same way every time
- Bad input must be refused, not silently scored

My responsibility

- Turn models into a tested, reproducible scoring pipeline
- Integrate the Data Science model
- Prove it works, and document what it can't do

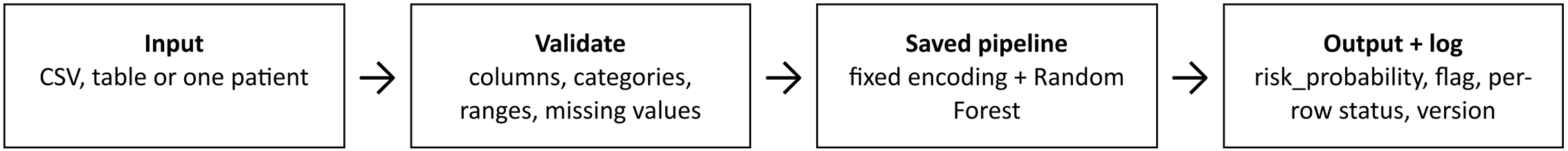
PROJECT JOURNEY

From notebook to integrated pipeline

Week	Stage	What I delivered
Week 6	Development	Data pipeline over 27 tables, 75 features, 6 models, each compared against a baseline. Only High-Risk Patient showed lift
Week 7	Testing → Refinement	Week 6 code passed **2/14** reliability checks → refined pipeline **14/14**; 61 automated tests
Week 8	Final integration	Reproduced on a second OS, error-handling matrix, end-to-end check, Data Science contract tests

MY TRACK CONTRIBUTION

What I built: the scoring pipeline



Methods & tools

- Python, pandas, scikit-learn Pipeline, joblib
- pytest + mutation testing
- Command-line tool with exit codes and logging

Design decisions

- Preprocessing saved inside the model, so it can't drift
- Reject impossible or missing input with the line and reason
- Threshold kept in metadata, adjustable without retraining

RESULTS

Evidence that it is ready to demonstrate

14/14
reliability checks
(Week 6 code: 2/14)

65/65
automated tests pass
5/5 injected bugs caught

12/12
error cases handled
as specified

Identical
retrain on Windows +
Python 3.12.10

High-Risk model, held-out test set (n = 1,250)	**Value**
ROC-AUC (Week 6 → Week 7, after removing leaked features)	0.727 → 0.721
Lift when the clinic flags its top 10% / 20% / 30% (replaces the 0.33 threshold)	1.77 / 1.82 / 1.84
Data Science release gate (AUC, 95% CI, shuffle test, lift)	Passed, but Age ≥ 65 alone: 0.721

Bad input is refused, with a reason

Input	Exit code	What the pipeline does
Valid patient	0	Scored
Age 250	2	Rejected: “CSV line 2, Age: 250 outside valid range [0, 120]”
Age blank / “fifty”	2	Rejected: missing value / not a number
VisitType = Surgery	2	Rejected, lists the allowed categories
Mixed batch with --skip-invalid	0	Good rows scored; bad rows marked REJECTED with the reason, and written to a rejects file
Age 110 (valid, outside training data)	0	Scored, with an extrapolation warning in the log
Model file missing	3	Clear error: run the training script first

CROSS-TRACK COLLABORATION

Data Science ↔ ML Engineering

Received from Data Science

- Final model package (admission-proxy scorer)
- Requirements note + 15 interface contract tests
- Status: **validated null result**, not for decisions

What I did with it

- Reviewed the code and the model file before running
- Ran the tests in two environments
- Ran the same tests against my pipeline
- Fed my data into their scorer

15/15
DS package in my environment
(Py 3.12 / sklearn 1.9.1 and 3.14 / 1.9.0)

11 → 15/15
DS tests against my pipeline
(7 Nov → shared contract, 8 Nov)

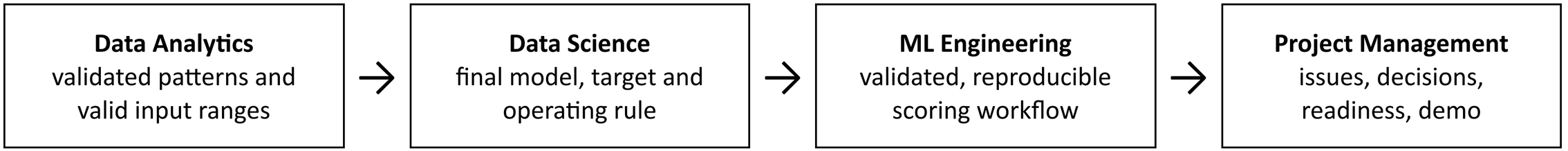
5,000
of my rows scored by the DS model,
0 rejected

CROSS-TRACK COLLABORATION

From two contracts to one

What the tests revealed	What we agreed and did
Contracts disagreed on missing values, unknown categories and empty input	One shared contract: reject with a reason, per-row status, empty in → empty out. Both sides changed: **15/15**
Age 110 vs 120; decimals vs whole numbers	Shared limits: Age 0–120, whole numbers
PriorVisits might count future visits	Confirmed point-in-time, same as mine
DS release gate applied to my High-Risk model	Passes (AUC 0.721), but Age ≥ 65 alone scores 0.721: it restates its own label → **demo only**
Threshold 0.33 flags two-thirds of patients	Flag budget: report lift at 10/20/30%; the clinic picks the percentage

How my work connects



- **Business value:** a safe, repeatable way to score patients, so the clinic can test prioritising reminders and follow-up for missed appointments once a model has real signal
- **My part:** any model Data Science produces can be scored the same way, with bad input refused and every run logged
- The GenAI assistant answers patient admin questions from the Knowledge Base; it does not call the model

LIMITATIONS

What the solution cannot do yet

- **No usable predictor:** the DS model has no signal (hold-out AUC 0.467); my High-Risk model passes the release gate only by restating the Age ≥ 65 part of its constructed label
- **Synthetic data:** only 5,000 of 60,000 appointments have full context
- **Flag budget not set:** the clinic must choose 10, 20 or 30% from its follow-up capacity
- **Not production:** command-line tool and Python API only; no web service, authentication or monitoring
- **Fairness:** province and facility-type checks not possible with the current data

Ready to demonstrate, not to make decisions about real patients.

CLOSING

Key takeaway and next steps

Takeaway

A model is only useful when others can run it safely and get the same answer. Testing showed the Week 6 code wasn't usable; it now is, and the evidence proves it.

What I learned

- Test the pipeline, not just the model
- Collaboration needs agreed contracts
- A release gate must be paired with a check of how the target is built

Future version

- Retrain on real outcomes (attendance, admissions)
- Clinic sets the flag budget
- Web API, monitoring and fairness checks

Thank you.